

TACT: Touch-free Acoustic Control Triggers for LLMs

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Current Large Language Model (LLM) voice interfaces often fall short in hands-free scenarios, requiring physical interaction or continuous active listening, which presents accessibility limitations in “hands-busy” environments and raises privacy concerns. This project presents TACT: Touch-free Acoustic Control Triggers, an interaction framework that uses non-verbal acoustic gestures, such as knocks, claps, and tongue clicks, to manage LLMs without physical touch. By leveraging Apple’s SoundAnalysis Classifier, the system maps distinct acoustic sounds to functional commands, such as pausing, adjusting speech rate, or switching models. Evaluations with 5 participants demonstrated a 100% task success rate during qualitative proficiency test. However, quantitative performance varied across gesture types: while waves (100%) and double knocks (90%) proved highly reliable, tongue clicks achieved only 40% accuracy due to significant anatomical variability among users. Our findings suggest the viability of acoustic gestures for privacy-preserving, hands-free LLM management, with future work focusing on fine-tuning specialized models like Google YAMNet to better accommodate various user-specific nuances.

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1 Introduction

While text-based input remains dominant with Large Language Models (LLMs), such as Gemini and ChatGPT, mobile applications of them feature voice interfaces that facilitate speech-based interaction. However, the control mechanisms for these interfaces predominantly rely on explicit tactile inputs or continuous active listening. This reliance imposes two primary burdens on user: physical constraints and privacy concerns. In “hands-busy, eye-busy” situations, ranging from cooking, driving, to feeling physical fatigue, manual interaction is often impractical or unsafe. Furthermore, the “always-listening” nature of these interfaces can trigger psychological discomfort, as users often feel apprehensive about their private environments being constantly monitored for valid voice triggers.

Another limitation of current interfaces is the restricted range of settings available during an active session. Currently, in-session control is mostly limited to basic input toggles, such as muting the microphone or sharing a screen. While services like Perplexity Voice and Claude Voice offer speech rate adjustments through one-depth settings (accessible directly within the session), Gemini Live and ChatGPT Voice still require users to navigate back to the root settings menu. Even with these options, none of these interfaces support mid-conversation output/response control, such as switching models or repeating the last response.

To address these limitations, this project proposes TACT: Touch-free Acoustic Control Triggers, a framework designed to manage voice assistant settings and actions via non-lexical, reproducible acoustic patterns. By transitioning the control mechanism from physical touch to discrete acoustic gestures, such as knocks or claps, TACT provides a privacy-conscious, hands-free alternative. This framework enables users to execute critical mid-conversation actions, including model switching, speech rate adjustments, and response repetition, thereby lowering the physical and psychological barriers inherent in existing LLM voice modes.

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2 Related Work

Prior research has extensively explored alternative ways to direct intelligent interfaces and managing dialogue flow. For instance, Chen et al. integrated physical gestures with audio-haptic feedback to help users navigate system interactions. While their approach shows how multiple sensory inputs can improve dialogue management, it still relies on direct tactile manipulation and often requires specialized haptic hardware. In contrast, TACT removes the need for contact entirely. By combining a non-contact proximity wave with discrete acoustic triggers, TACT makes interaction possible under "hands-busy" constraints where touching a device is unfeasible or hazardous.

Other researchers have focused on the meaning of non-verbal sounds and how they can address privacy concerns. Bevilacqua et al. defined a framework for "sonic gestures," which helps us understand how non-speech acoustic patterns can carry specific user intent. Regarding privacy, Zeng et al. demonstrated how cues like gaze and voice volume can act as controls for smart speakers, offering an alternative to "always-listening" states. While Bevilacqua provides the theory and Zeng addresses the discomfort of constant monitoring, their actual implementations are mostly limited to theoretical models or simple hardware toggles. TACT differs by applying these sonic gestures to a practical framework for complex, mid-conversation LLM adjustments, such as changing models or speech rate, rather than just toggling a microphone.

Crucially, our methodology is most directly inspired by Talon Voice and its `parrot.py` integration. Talon Voice is a powerful hands-free computing platform that trains neural networks to recognize discrete acoustic noises, such as clicks, pops, or hisses, to execute arbitrary computer commands. While TACT uses a similar logic, the two systems have very different goals. Talon is primarily an accessibility tool meant to replace traditional keyboards, typically requiring high-quality microphones. Conversely, TACT is built for mobile smartphones. This means the system must function reliably despite lower-quality mobile mics and environmental noise, serving as a way to manage the live state of an LLM conversation rather than acting as a direct replacement for general computer commands.

3 Implementation & Experiment Details

Because of iOS restrictions regarding background microphone access and the challenges of direct integration with commercial LLM interfaces, we developed TACT as a standalone mobile application rather than a background service. This prototype effectively simulates a live voice interface, enabling us to test mid-conversation interactions in a controlled environment without the overhead of a fully backgrounded LLM. The interaction logic for this system is outlined in Figure 1.

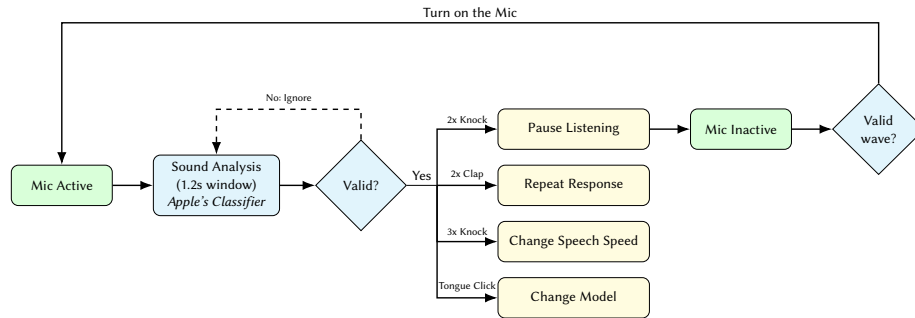


Fig. 1. TACT System Design: Logic flow for acoustic trigger processing and model state management.

Technical Implementation

On the technical side, the architecture relies on Apple’s SoundAnalysis framework, a built-in neural network optimized for the Apple Neural Engine that can classify approximately 300 acoustic categories on-device. The audio pipeline processes incoming microphone data through a 1.2-second sliding analysis window, operating with a 250ms hop size (a 0.75 overlap). To filter out false positives caused by sounds people naturally produce while speaking, we implemented a system gate that suppresses gesture detection whenever the user is actively speaking.

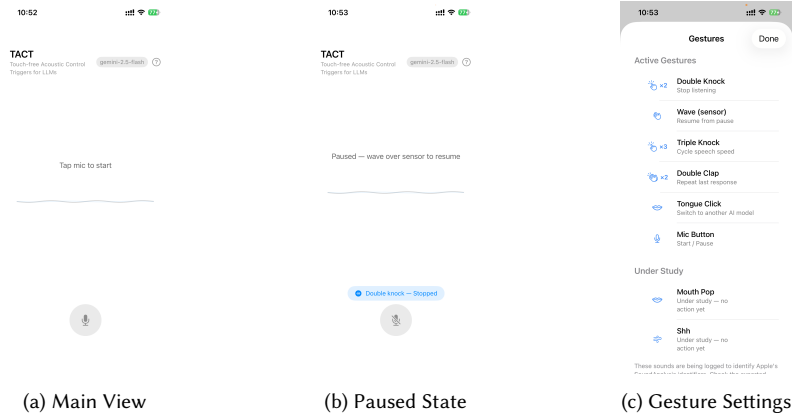


Fig. 2. Primary TACT interfaces.

While Apple’s SoundAnalysis framework outputs a raw vector of confidence scores per category, we constrained our trigger logic to the top five categories to maintain real-time responsiveness. Drawing on the goal of mapping distinct acoustic signals to system actions, we iteratively tuned our classification strategies to balance sensitivity with false-positive prevention. These thresholds were refined through multiple rounds of pilot testing involving two primary testers (one male, one female) and six independent one-time testers.

- **Double Knock:** Used to pause listening. It is evaluated using a single-label strategy focused on the primary knock identifier with a confidence threshold of ≥ 0.50 and a 0.5-second debounce to prevent accidental re-triggering.
- **Triple Knock:** Used to cycle through speech rates, this gesture required a specialized temporal windowing approach. During initial development, we observed that the third knock in a sequence frequently registered with lower acoustic intensity than the first two. To accommodate this diminishing signal, we implemented a 1.5-second observation window that allows the system to aggregate the sequence even when the final knock falls near the noise floor.
- **Wave (over sensor):** Acting as a hardware-level gate, the proximity wave resumes listening when the microphone is inactive. By utilizing the device’s proximity sensor, this provides a strictly non-acoustic wake trigger for scenarios where the mic was manually muted or toggled off via a double knock.
- **Double Clap:** To account for the high variance in clapping styles and surface resonance, we moved beyond a single-label approach in favor of a composite strategy combining 8 percussive labels. The system requires a total percussive score ≥ 0.45 paired with a specific clap signal score ≥ 0.10 . An extended 0.8-second debounce was added to ensure that the resonant “echo” of a single physical clap was not miscounted as a double-trigger.

- **Tongue Click:** This gesture switches the AI model (e.g., between Gemini Flash 2.5 and Gemini Flash 2.5 Lite). To ensure robustness during active dialogue, we implemented a composite score of mouth-related sounds (≥ 0.35) coupled with a negative speech constraint (< 0.50). This logic effectively suppresses false triggers that might otherwise occur during regular conversation.

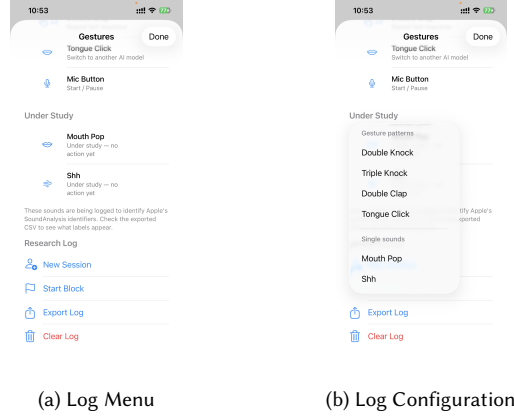


Fig. 3. Logging and data collection interfaces.

3.0.1 Data Logging. To help evaluate the system and fine-tune the thresholds, we built a logging system directly into the application. Whenever an acoustic event is triggered, the app records a timestamped row that includes the gesture name and the top five classifications and confidence scores provided by the Apple classifier (formatted as `timestamp, fired_as, top1, conf1, etc.`).

Since the app processes audio every 250ms, the logs often capture sounds that the user did not intend to be commands. For example, the system might pick up a breath that sounds like a whistle or a slight mouth noise immediately after a clap. Because these unintentional sounds appear in the logs, we could not rely on the raw data alone to calculate accuracy. Instead, a researcher moderated every test session, manually recording what sounds the user actually made. By comparing these human-recorded notes with the app’s automated logs, we were able to distinguish between actual system errors and random background noise. This approach ensured that our final accuracy results reflected how well TACT actually understood the user’s intentional commands.

Experiment Details

We evaluated TACT with five participants (three female, two male) in their twenties. This small cohort was chosen to capture a range of acoustic diversity; for example, one participant’s bony hand structure resulted in much lower resonance during knocks, while another produced exceptionally sharp, high-pitched claps. We also observed significant variability in tongue clicks. One participant generated a thick, low-frequency sound that shared spectral similarities with a hammer hit, causing the baseline classifier to misidentify it.

The evaluation was split into two primary phases. The first phase was a qualitative proficiency task designed to see how quickly users could learn and apply the gestures in a real scenario. After a brief onboarding period of less than two minutes, participants were asked to use the Gemini voice interface to find out how to write the word “Congratulations” in Spanish. This task required them to use at least three different TACT gestures. To simulate real-world stress, the

researcher intentionally interrupted the participants by calling their names mid-conversation. This forced the users to react dynamically, using a double knock to pause the AI or a double clap to ask the model to repeat what was missed during the interruption.

The second phase was a quantitative reliability test to measure the raw accuracy of the triggers. Each participant performed six attempts for each gesture type, resulting in 30 trials per gesture across the group. For the tongue click specifically, one participant (P4) completed only five trials due to a session recording error, which was offset by a seventh trial from another participant (P5) to maintain the total sample size of 30. To ensure we did not count accidental background noise as system failures, a moderator observed each session to record the user’s actual intent. We then cross-referenced these moderator notes with the raw system logs to calculate final accuracy rates and identify the specific “boundary cases” where individual physical differences made detection difficult.

4 Results

The evaluation demonstrated strong overall performance in terms of user proficiency. In the qualitative task, all five participants successfully completed the interactive scenario, achieving a 100% task success rate (5/5) with an average onboarding time of less than two minutes. This suggests that the acoustic gesture mappings are intuitive and require minimal training for basic use.

However, we observed a slight learning curve related to individual acoustic signatures. One participant(P4) initially experienced friction during the onboarding phase because his natural tongue click style fell outside the system’s baseline classification threshold. This lack of immediate feedback caused temporary confusion regarding how the system was interpreting his input. Despite this initial hurdle, the participant was able to successfully adapt his technique after a few attempts, ultimately completing the task. This observation highlights both a potential point of failure for users with unique acoustic profiles and the system’s inherent learnability as users naturally calibrate their gestures to the interface.

The quantitative evaluation measured the raw reliability of each trigger across 30 total trials per gesture ($N = 5$). The detection accuracy results were as follows:

- **Wave:** 100% (30/30)
- **Double Knock:** 90% (27/30)
- **Triple Knock:** 87% (26/30)
- **Double Clap:** 70% (21/30)
- **Tongue Click:** 40% (12/30)

As illustrated in Figure 4, there is a distinct performance disparity between surface-based gestures, such as Wave, Double Knock, and Triple Knock, and sounds produced by the human body, such as Double Clap and Tongue Click. The Wave gesture achieved a 100% success rate. Because it utilizes the iPhone’s proximity sensor rather than acoustic classification, it remains entirely resilient to ambient noise and individual user differences. The knock-based gestures both exceeded 85% accuracy, largely because striking a hard surface like a desk or a wall creates a dominant, consistent knock identifier within the Apple classifier. In our testing, we observed confidence scores for these knocks ranging from 0.628 to 1.000 across all participants. Conversely, the Double Clap and Tongue Click performed significantly worse, suggesting that gestures involving human anatomy are far more susceptible to physical variability.

Qualitative observations confirmed that anatomical and stylistic variance were the primary drivers of detection failure. For instance, the bony hand structure of P2 produced significantly lower resonance during knocks and the participant had a difficulty producing consistent tongue clicks difficult to execute. P4 generated thick, low-frequency

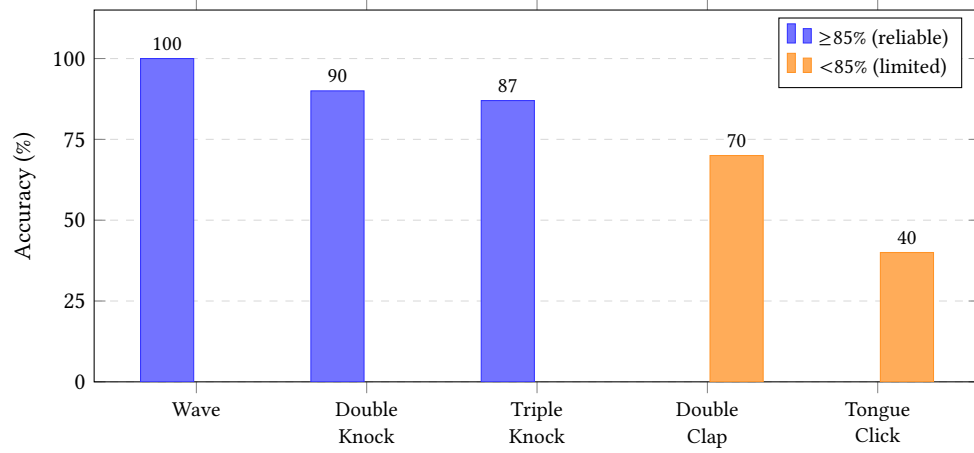


Fig. 4. Gesture detection accuracy across $N = 5$ participants ($n = 30$ trials per gesture). Blue bars denote gestures that reached $\geq 85\%$ accuracy; orange bars denote gestures with higher inter-participant variance.

Table 1. Per-participant detection accuracy ($n = 6$ trials each, except P4 Tongue Click where $n = 5$ and P5 Tongue Click where $n = 7$). Cells below 70% are highlighted.

Gesture	P1	P2	P3	P4	P5	Total
Wave	6/6 (100%)	6/6 (100%)	6/6 (100%)	6/6 (100%)	6/6 (100%)	30/30 (100%)
Double Knock	6/6 (100%)	4/6 (67%)	5/6 (83%)	6/6 (100%)	6/6 (100%)	27/30 (90%)
Triple Knock	6/6 (100%)	5/6 (83%)	4/6 (67%)	6/6 (100%)	5/6 (83%)	26/30 (87%)
Double Clap	4/6 (67%)	4/6 (67%)	3/6 (50%)	5/6 (83%)	5/6 (83%)	21/30 (70%)
Tongue Click	2/6 (33%)	0/6 (0%)	4/6 (67%)	0/5 (0%)	6/7 (86%)	12/30 (40%)

tongue clicks that the classifier frequently misidentified as a knock or hammer strike due to spectral similarity. Similarly, the sharp claps produced by P3 generated strong `finger_snapping` and `tap` signals that overlapped with other gesture categories, leading to less reliable double-clap detection. P5, who produced the most distinct and clear tongue clicks, was the only participant to exceed 80% accuracy for that specific gesture.

Beyond these inter-participant differences, intra-participant variability also contributed to classification instability. We observed that a single user rarely produces an identical acoustic signal across multiple attempts. For example, clapping with a full palm generates a broad, low-frequency impact dominated by `slap_smack` and `hammer` labels, whereas clapping with fingers against the palm produces a sharper, high-frequency signal closer to `tap` or `finger_snapping`. This lack of uniform execution means that even if a user's average profile passes the detection threshold, individual "outlier" attempts often fall short. This explains why the overall double-clap accuracy remained at 70%, despite some participants individually reaching 83% in controlled trials.

A final significant finding involves the biomechanical pattern of gesture repetition. During triple-gesture attempts (e.g., triple knock), the third strike was consistently softer than the preceding two. This decay in intensity appears to be a natural pattern: the initial strikes are delivered with deliberate force, while the third follows with reduced physical energy. Consequently, the third event frequently produced confidence scores near or below the detection threshold,

causing the system to register only two events and route the user to the wrong action. This effect was notably more pronounced for claps than knocks. Since clap detection relies on a composite score across multiple labels, there was a narrower margin for a weakened third signal to clear all required conditions simultaneously. Therefore, we excluded triple clap from the gesture set.

Sound Detection Strategy and Threshold Analysis

A significant technical takeaway from this project is the necessity of varied detection strategies based on the acoustic profile of the gesture. Because Apple’s SoundAnalysis .version1 classifier lacks a one-to-one mapping for every gesture in our library, we implemented two distinct approaches: single-label detection and composite scoring.

For knocks, the knock identifier proved robust enough for a single-label strategy. We set the confidence threshold at $\text{knockConf} \geq 0.50$, an increase from our initial 0.45 baseline. This change was prompted by observations that certain clapping styles occasionally generated knock scores as high as 0.458, leading to misclassifications. While this adjustment sacrificed sensitivity for some "edge-case" knocks, such as one recorded at 0.484, it significantly improved the system’s overall precision by reducing false positives.

The detection of claps required a more nuanced approach, as no single label consistently dominated the output. Instead, we calculated an aggregated percussive score by summing confidence levels across eight acoustically related categories: hammer, slap, tap, thump, thud, clap, applause, and chopping. This strategy accounts for how the system distributes "acoustic energy" across multiple labels. For example, a single clap might register as a mix of hammer (0.333), clapping (0.283), and tap (0.145). While none of these individual scores would trigger an action, their sum (1.073) is clearly over our 0.45 composite threshold. We also implemented a secondary guard requiring a minimum "clap signal" score of 0.10 (combining clapping, slap, and applause) to ensure the sound was indeed a clap and not a generic percussive noise.

For tongue clicks, we faced a unique challenge as the classifier lacks a dedicated category. We derived a `tongueScore` using `click`, `tick`, and `tongue` as proxy identifiers. Based on session logs, we lowered the threshold from 0.40 to 0.35 to capture genuine clicks that previously went undetected at the 0.377 mark. Despite this tuning, inter-participant variance remained a hurdle. While P5 produced reliably detectable `click` scores ranging from 0.363 to 0.581, the clicks from P4 never exceeded 0.174. In the latter case, the classifier’s inability to distinguish the user’s specific acoustic signature resulted in the sound being mislabeled as speech or hammer, highlighting the limitations of a general-purpose classifier for highly individualized body sounds.

Sensitivity and False-Positive Trade-off

A consistent design trade-off emerged across all gesture types: lowering a threshold to capture softer or more varied sounds proportionally increased the risk of false positives from background noise or other gestures. Table 2 summarizes the final thresholds and debounce settings used in the study. While we initially experimented with additional gestures like "mouth pops" or "shh" sounds, these were excluded for the formal reliability tests due to low baseline accuracy. However, they remain promising candidates for future work using more specialized acoustic models.

The temporal logic of the system also introduced specific interaction constraints. The analysis window fires every 250 ms (using a 1-second window with a 0.75 overlap factor), which means a single physical sound can often span two or three consecutive windows. To prevent a single gesture from being counted multiple times, we had to implement precise debounce intervals. For example, we increased the clap debounce to 0.8 seconds after observing that single claps were frequently being registered twice and misidentified as double-claps. While this solved the double-counting

Table 2. Final detection thresholds and debounce values per sound type, with tuning direction relative to initial values.

Sound	Strategy	Threshold	Debounce
Knock	Single label	≥ 0.50	0.5s
Clap	Composite (8 labels)	$\geq 0.45 + \text{signal} \geq 0.10$	0.8s
Tongue Click	Composite (3 labels)	≥ 0.35	0.5s
Mouth Pop	Single label (snapping)	≥ 0.75	0.5s
Shh	Composite (hiss/noise)	≥ 0.30	1.0s

problem, it imposed a strict timing requirement on the user: if a participant claps faster than the 0.8-second limit, the second clap is ignored. This represents the fundamental tension between sensitivity and reliability that guided all of our threshold and timing decisions throughout the project.

5 Discussion, Limitation & Conclusion

The growing adoption of LLM voice modes has introduced a new interaction paradigm, yet one that remains fundamentally constrained by its reliance on tactile and continuous vocal input. Current implementations still require physical taps to manage sessions, rely on "always-on" microphones that spark privacy concerns in shared spaces, and offer very little in-session configurability. TACT proposes a shift in this framework by introducing an acoustic gesture layer alongside speech. This allows users to manage LLM sessions through non-verbal sounds when physical interaction is impossible or impractical, all without the privacy risks associated with constant environmental monitoring.

Our results validate the feasibility of this framework for a specific subset of interaction modalities. The success of the Wave (100%), Double Knock (90%), and Triple Knock (87%) gestures demonstrates that surface-contact and proximity-based triggers are reliable. The fact that all five participants completed the interactive task with less than two minutes of onboarding suggests that the gesture vocabulary is intuitive and can support genuine "hands-busy" scenarios where traditional touch interaction would fail.

Crucially, TACT is designed to complement existing commercial infrastructures. Since platforms like Perplexity Voice and Claude Voice operate as speech-to-text and text-to-speech pipelines surrounding a core language model, they are not natively multimodal. TACT shares this architecture, meaning a gesture control layer could be integrated into these platforms with minimal structural friction. Acoustic gestures offer a path to surfacing these controls "inline," allowing users to adjust settings without breaking the conversational flow.

However, the evaluation also exposes the clear boundaries of this approach. The performance of the Double Clap (70%) and Tongue Click (40%) reveals that body-produced sounds introduce a level of variance that a fixed, general-purpose classifier cannot yet accommodate. The fact that two participants scored 0% on the Tongue Click is particularly telling. These were not failures of system design, but rather a limitation of sound variance and the Apple SoundAnalysis .version1 framework, which lacks a dedicated category for tongue clicks and uses fixed decision boundaries that cannot be adapted to individual acoustic profiles.

Because the underlying model's training data and internal logic are undisclosed, debugging specific misclassifications is difficult. Furthermore, empirical threshold tuning that works for one participant may fail entirely for another due to different anatomical resonances. These trade-offs create a persistent tension: every threshold adjustment that improves recall for softer sounds increases the risk of ambient noise triggering a false action, while every debounce increase

to prevent double-counting restricts how quickly a user can perform a gesture. Ultimately, these are not incidental bugs, but inherent tensions that arise when repurposing a general-purpose audio scene classifier as a discrete gesture recognition engine.

In conclusion, TACT demonstrates that acoustic gestures offer a practical, privacy-conscious alternative for managing LLM voice modes, especially in scenarios where physical touch is impossible. Its compatibility with existing voice architectures makes it a viable candidate for integration into current commercial platforms, addressing key control gaps that remain unresolved today. While the system is intuitive and easy to learn, its current reliability is limited by the static nature of the underlying classifier, which lacks the flexibility to adapt to individual acoustic differences.

The future work of this research will focus on replacing Apple’s .version1 framework with Google’s YAMNet, which supports transfer learning and fine-tuning. This shift would allow decision boundaries to be tailored to each gesture class using a larger, more diverse participant dataset. Additionally, making the gesture-to-action mapping user-configurable would allow individuals who cannot reliably produce specific sounds to substitute alternative triggers, thereby improving the system’s accessibility. Together, these advancements would move TACT from a fixed-gesture prototype toward an adaptive, user-calibrated interaction layer specifically optimized for the nuances of LLM voice interfaces.

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